ACC3018 Applied Analytics Capstone

Seminar 2

Research Methods

- Introduction
- Summary Statistics
- Correlation & Covariance
- Hypothesis Testing (t-test)
- Linear Regression:
  - Time Series/Cross-Section
  - Panel Data/Fixed Effects
  - Difference-in-difference (DiD)
  - Sub-sample analysis
- Logit Regression
- Fama-Macbeth Regression
- Survey Data Analysis

Summary Statistics : Sample Mean

Mean is defined as the numerical average, calculated as the sum of the data values divided by the number of values.

$$x_1, x_2, \ldots, x_N$$

Sample mean, denoted as $\bar{x}$, is defined as:

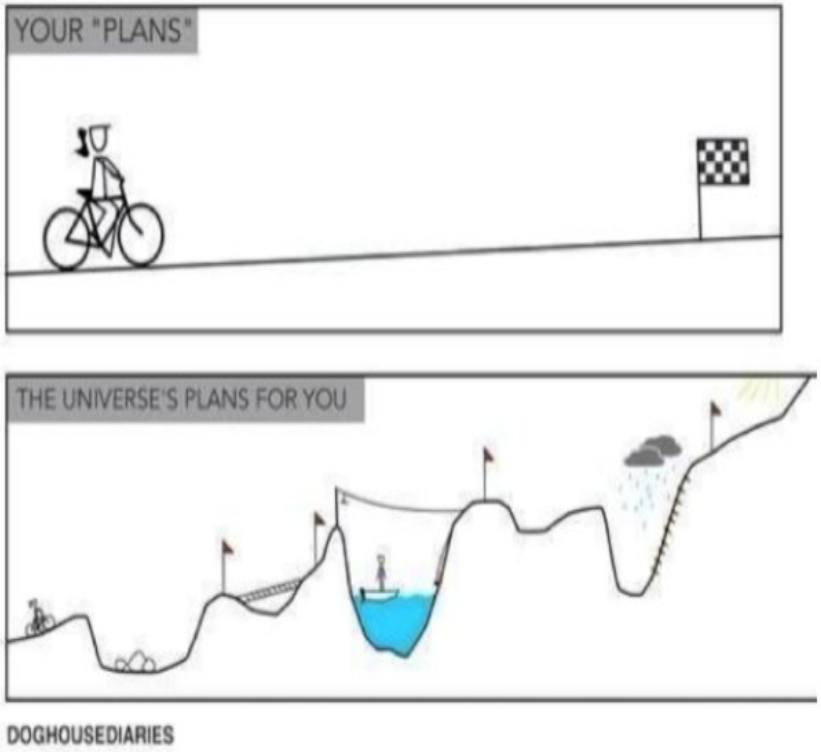
$$\boxed{\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\sum x_i}{n}}$$

where $x_1, x_2, \ldots, x_n$ is a subset of $n < N$ data in the population.

Summary Statistics : Sample Mean and Sample SD

Medallion Performance Data: 1998 to 2018

Year	Medallion Gross Return	Fixed Fee	Performance Fee	Medallion Net Return	Medallion fund size (millions)
1988	16.30%	5%	20%	9.04%	20
1989	1.00%	5%	20%	-3.20%	20
1990	77.80%	5%	20%	58.24%	30
1991	54.30%	5%	20%	39.44%	42
1992	47.00%	5%	20%	33.60%	74
1993	53.90%	5%	20%	39.12%	122
1994	93.40%	5%	20%	70.72%	276
1995	52.90%	5%	20%	38.32%	462
1996	44.40%	5%	20%	31.52%	637
1997	31.50%	5%	20%	21.20%	829
1998	57.10%	5%	20%	41.68%	1,100
1999	35.60%	5%	20%	24.48%	1,540
2000	128.10%	5%	20%	98.48%	1,900
2001	56.60%	5%	36%	33.02%	3,800
2002	51.10%	5%	44%	25.82%	5,240
2003	44.10%	5%	44%	21.90%	5,090
2004	49.50%	5%	44%	24.92%	5,200
2005	57.70%	5%	44%	29.51%	5,200
2006	84.10%	5%	44%	44.30%	5,200
2007	136.10%	5%	44%	73.42%	5,200
2008	152.10%	5%	44%	82.38%	5,200
2009	74.60%	5%	44%	38.98%	5,200
2010	57.50%	5%	44%	29.40%	10,000
2011	71.10%	5%	44%	37.02%	10,000
2012	56.80%	5%	44%	29.01%	10,000
2013	88.80%	5%	44%	46.93%	10,000
2014	75.00%	5%	44%	39.20%	9,500
2015	69.30%	5%	44%	36.01%	9,500
2016	68.60%	5%	44%	35.62%	9,500
2017	85.40%	5%	44%	45.02%	10,000
2018	76.40%	5%	44%	39.98%	10,000
Average	66.07%			39.20%	
St Dev	31.66%			20.34%	



Summary Statistics : Sample Mean and Sample SD

Panel A: Descriptive Statistics for U.S. Natural Disasters 1900 to 2010

Disaster Type	N	Mean No. of Fatalities	Mean Fatalities per Capita	Mean Economic Damage (2009$, Mil.)
Earthquake	806	4.7455	0.0605	87.677
Hurricane	9,665	2.4280	0.0482	17.376
Severe Weather	237,930	0.0581	0.0004	0.400
Urban Fire	2,466	6.0815	0.0164	10.295
Volcano	9	0.5556	0.1370	17.742
Weather	404,837	0.0490	0.0003	0.505
Wild Fire	2,206	0.0633	0.0002	7.949
All	657,919	0. 115	0.001	0.884

Panel B: Time-Invariant Disaster-Related CEO Characteristics

CEO-Specific Variables	N	Mean	Std. Dev.
No Fatality Experience	1,508	0.332	0.471
Medium Fatality Experience	1,508	0.558	0.497
Extreme Fatality Experience	1,508	0.110	0.314
No. of Disasters Experienced	1,508	25.867	41.86
No. of Fatal Disasters Experienced	1,508	3.906	5.408
Total Fatalities Experienced	1,508	15.708	38.515
Fatalities per Capita (%)	1,508	0.027	0.037
Economic Damage Experienced ($M)	1,508	0.216	1.881
Economic Damage per Square Mile ($M)	1,508	0.012	0.591
Average Fatality Risk of County 1900 to 2010	1,508	0.032	0.211
Average Econ. Damage Risk of County 1900 to 2010	1,508	0.301	1.229

Most events in our sample are weather-related (e.g., floods, thunderstorms, hail, etc.) and tend to be inconsequential in terms of human injuries and economic damages. In contrast, less frequent events such as earthquakes, fires, and hurricanes tend to have the direst consequences, both in terms of fatalities and economic damages.

Standard Deviation

Definition: Let $x_1, x_2, \ldots, x_n$ be the n data, and
$f_1, f_2, \ldots, f_n$ be the respective frequency

Sample variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2 f_i}{n-1}$$

$$s^2 = \frac{1}{n-1}\left[\sum x_i^2 f_i - \frac{\left(\sum x_i f_i\right)^2}{n}\right]$$

Sample Std. Dev: s

Correlation Table

Definition: Correlation is a statistical measure that expresses the extent to which two variables are linearly related (meaning they change together at a constant rate).

Sample correlation:

$$r = \frac{1}{n-1}\sum\left(\frac{x-\bar{x}}{s_x}\right)\left(\frac{y-\bar{y}}{s_y}\right)$$

	Sales/A	IBEI/A	CFO/A	Accruals/A	PROD/A	DISEXP/A	Abnormal CFO	Abnormal PROD	Abnormal DISEXP
IBEI/A	0.22								
CFO/A	0.11	0.71							
Accruals/A	0.18	0.57	−0.17						
PROD/A	0.95	0.13	0.01	0.17					
DISEXP/A	0.39	−0.16	−0.18	−0.01	0.15				
Abnormal CFO	−0.01	0.46	0.74	−0.22	−0.10	−0.10			
Abnormal PROD	−0.02	−0.22	−0.28	0.02	0.22	−0.48	−0.35		
Abnormal DISEXP	0.11	−0.08	−0.06	−0.04	−0.06	0.66	−0.17	−0.63	
Abnormal accruals	0.04	0.42	−0.18	0.81	0.04	−0.05	−0.22	0.03	−0.11

Hypothesis Test for the Mean

- Convert sample statistic ($\overline{X}$) to a Z_{STAT} or T_{STAT} test statistic.
- Assumptions: X follows Normal distribution or sample size is larger than 30.

Hypothesis Tests for μ

σ Known (Z test)

The test statistic is:

$$Z_{STAT} = \frac{\overline{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

σ Unknown (t test)

The test statistic is:

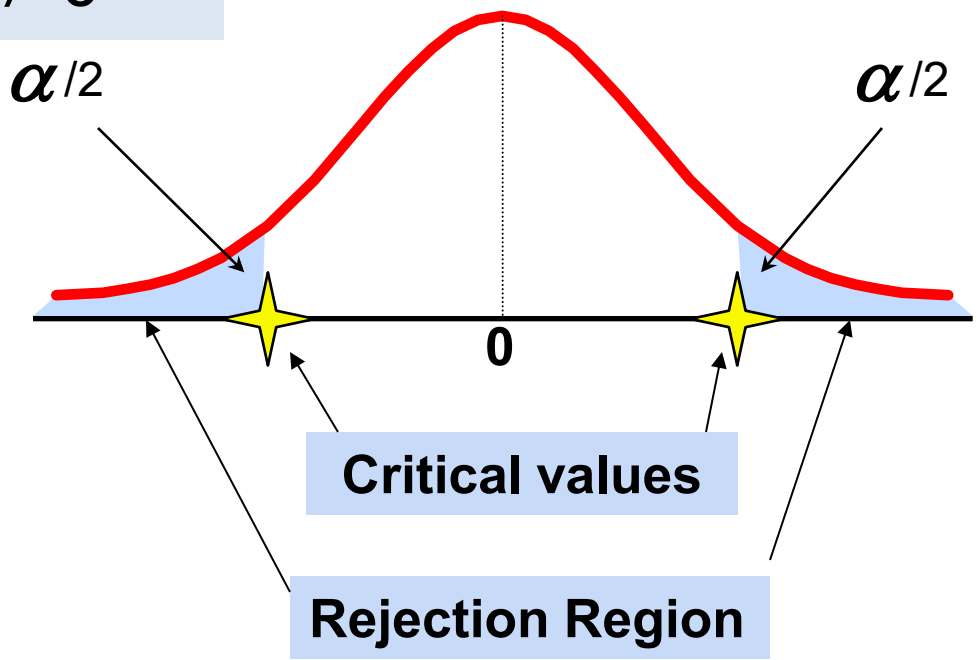
$$t_{STAT} = \frac{\overline{X} - \mu}{\frac{S}{\sqrt{n}}}$$

Hypothesis Testing

H_0: $\mu = 0$

H_1: $\mu \neq 0$

Level of significance = α



How do we decide the true mean, and the level of significance?

This is a two-tail test because there is a rejection region in both tails

Level of Significance

- If $p > 0.1$, then there will be no assumption for the null hypothesis
- If $p > 0.05$ and $p \leq 0.1$, it means that there will be a low assumption for the null hypothesis.
- If $p > 0.01$ and $p \leq 0.05$, then there must be a strong assumption about the null hypothesis.
- If $p \leq 0.01$, then a very strong assumption about the null hypothesis is indicated.

If we obtain a p-value equal to 0.03, then it indicates that there are just 3% chances of getting a difference larger than that in our research, given that the null hypothesis exists.

Hypothesis Testing

- Sample 1: mean $\overline{X}_1$
- Sample 2: mean $\overline{X}_2$
- Difference between two samples:

$$t = \frac{\Delta\overline{X}}{s_{\Delta\bar{X}}} = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{s^2_{\bar{X}_1} + s^2_{\bar{X}_2}}}$$

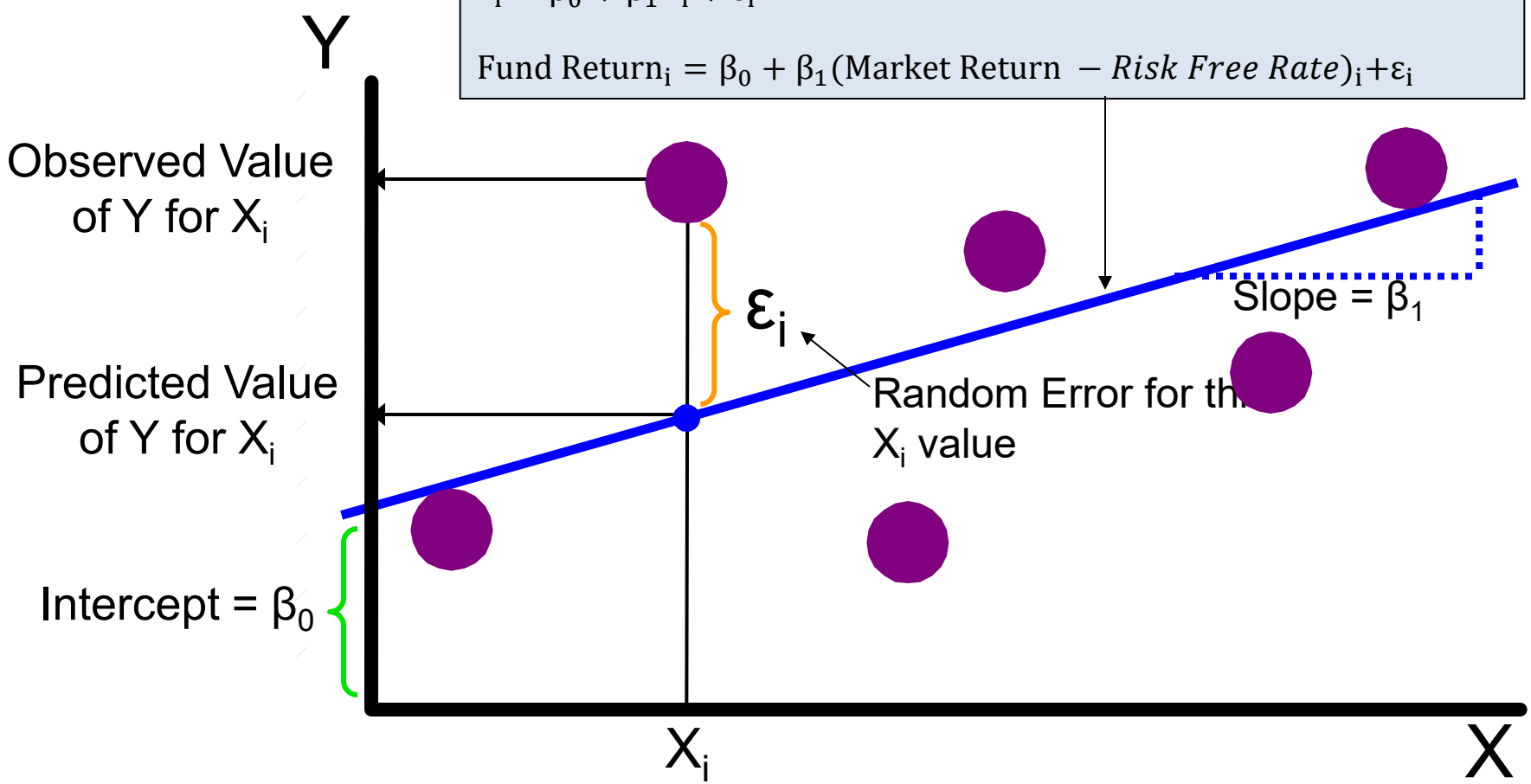
*Significant at the 1% level.
**Significant at the 5% level.

	Suspect firm-years		Rest of the sample		Difference in
	Mean	Median	Mean	Median	Means (*t*-stat)
Full sample of 21,758 firm-years with 503 suspect firm-years					
MVE ($ million)	745.82	75.60	1414.43	137.34	−668.61** (−5.84)
MVE/BVE	1.60	1.21	2.75	1.93	−1.15** (−10.64)
Total assets ($ million)	1180.57	153.17	1124.17	164.54	56.04 (0.34)
Sales ($ million)	1254.14	214.88	1394.35	221.05	−140.21 (−0.88)
IBEI ($ million)	2.81	0.29	61.80	4.46	−58.99** (−36.29)
CFO ($ million)	81.06	5.15	126.55	10.76	−45.49** (−3.45)
Accruals ($ million)	−78.24	−4.86	−64.67	−5.41	−13.57 (−1.07)
Sales/A	1.39	1.25	1.48	1.30	−0.09* (−1.92)
IBEI/A (%)	0.24	0.22	0.31	4.09	−0.07 (−0.53)
CFO/A (%)	4.54	4.77	6.50	8.25	−1.96** (−5.11)
Accruals/A (%)	−4.31	−4.54	−6.16	−5.20	1.85** (4.99)
Production costs/A (%)	98.99	80.45	97.08	78.79	1.91 (0.08)
Discretionary expenses/A (%)	36.63	30.31	44.16	37.44	−7.53** (−6.41)
Inventory turnover ratio	10.75	4.29	12.80	4.55	−2.05* (−1.88)
Receivables turnover ratio	19.55	6.00	17.40	6.10	2.15 (0.64)
Sub-sample of 17,338 firm-years, including 389 suspect firm-years					
Institutional ownership	31.11	26.87	35.32	33.26	−4.32** (−3.56)

Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\text{Fund Return}_i = \beta_0 + \beta_1(\text{Market Return} - \textit{Risk Free Rate})_i + \varepsilon_i$$



The Least Squares Method

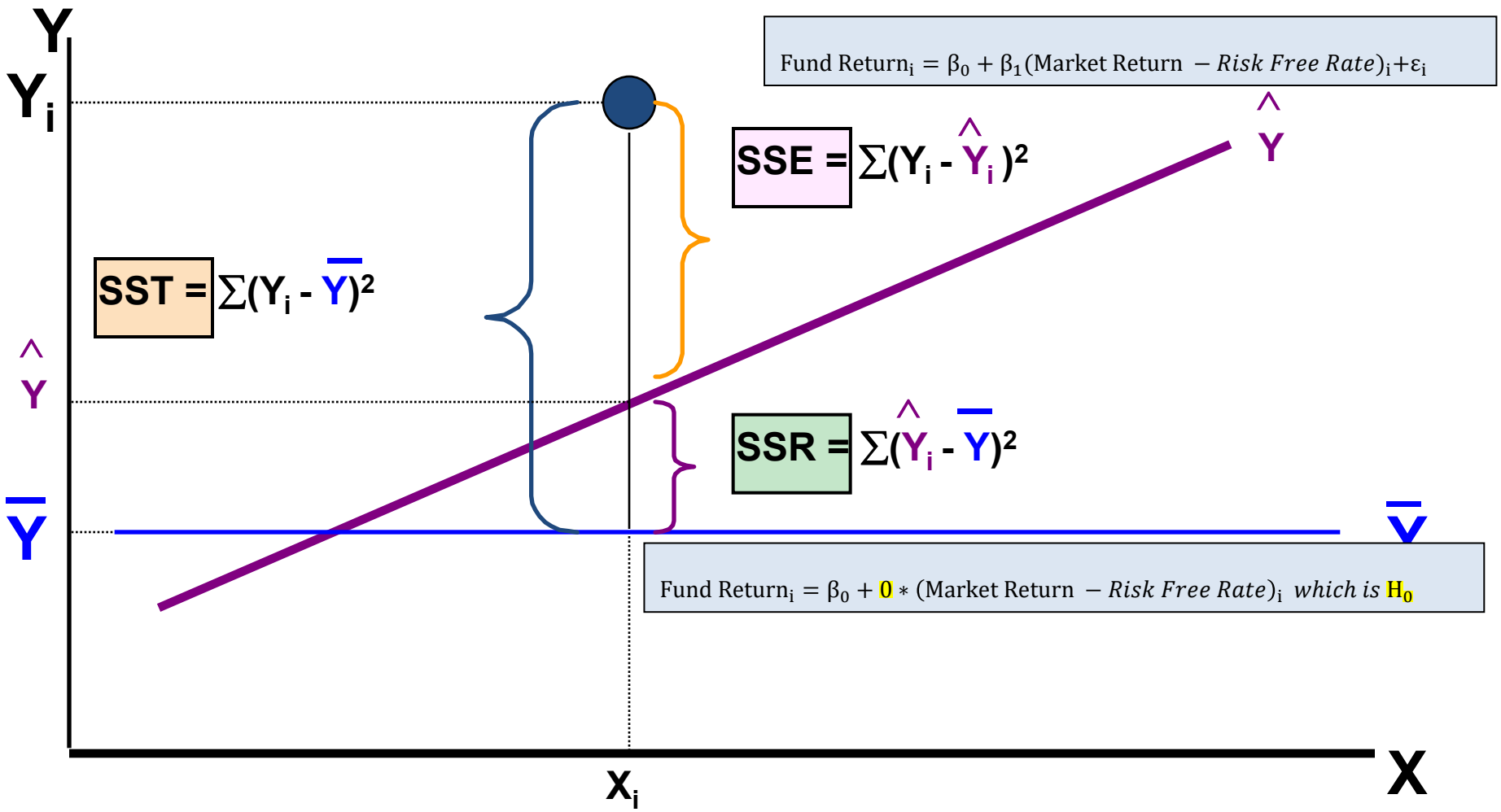
- The most common approach is **Ordinary Least Squares (OLS)**, which provide a "best fit" line that **minimizes the sum of squared residuals** of the linear regression model.
- More formally, **β_0 (intercept)** and **β_1 (slope)** solve the following minimization problem:

$$Find \min Q(\beta_0, \beta_1) \quad for\ Q(\beta_0, \beta_1) = \sum_{i=1}^{n} \varepsilon_i{}^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

- The optimal parameters are calculated to be:

$$\widehat{\beta_1} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \qquad \widehat{\beta_0} = \bar{y} - \widehat{\beta_1}\bar{x}$$

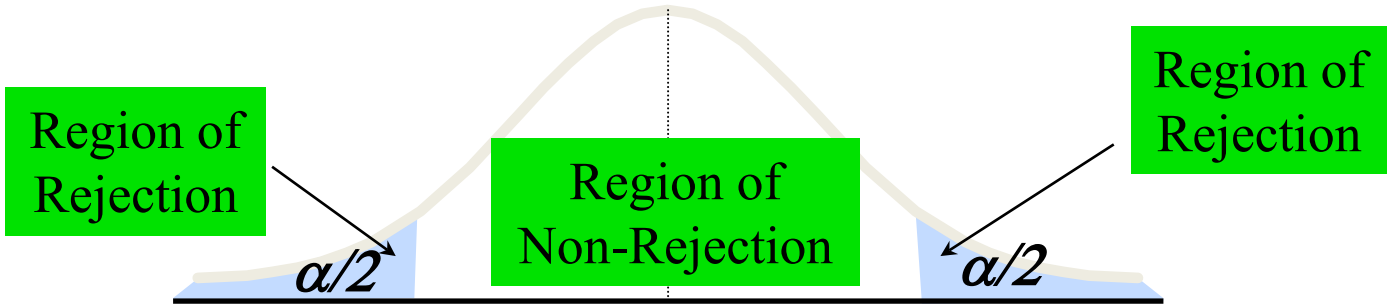
Simple Linear Regression



Simple Linear Regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Two-tail hypothesis testing for the slope: $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$



The slope is statistically significantly different from zero if the t-stat falls into the rejection region. Results interpretation: X can (or cannot) explain/predict Y

Research question: if the fund return generated from the market excess return?

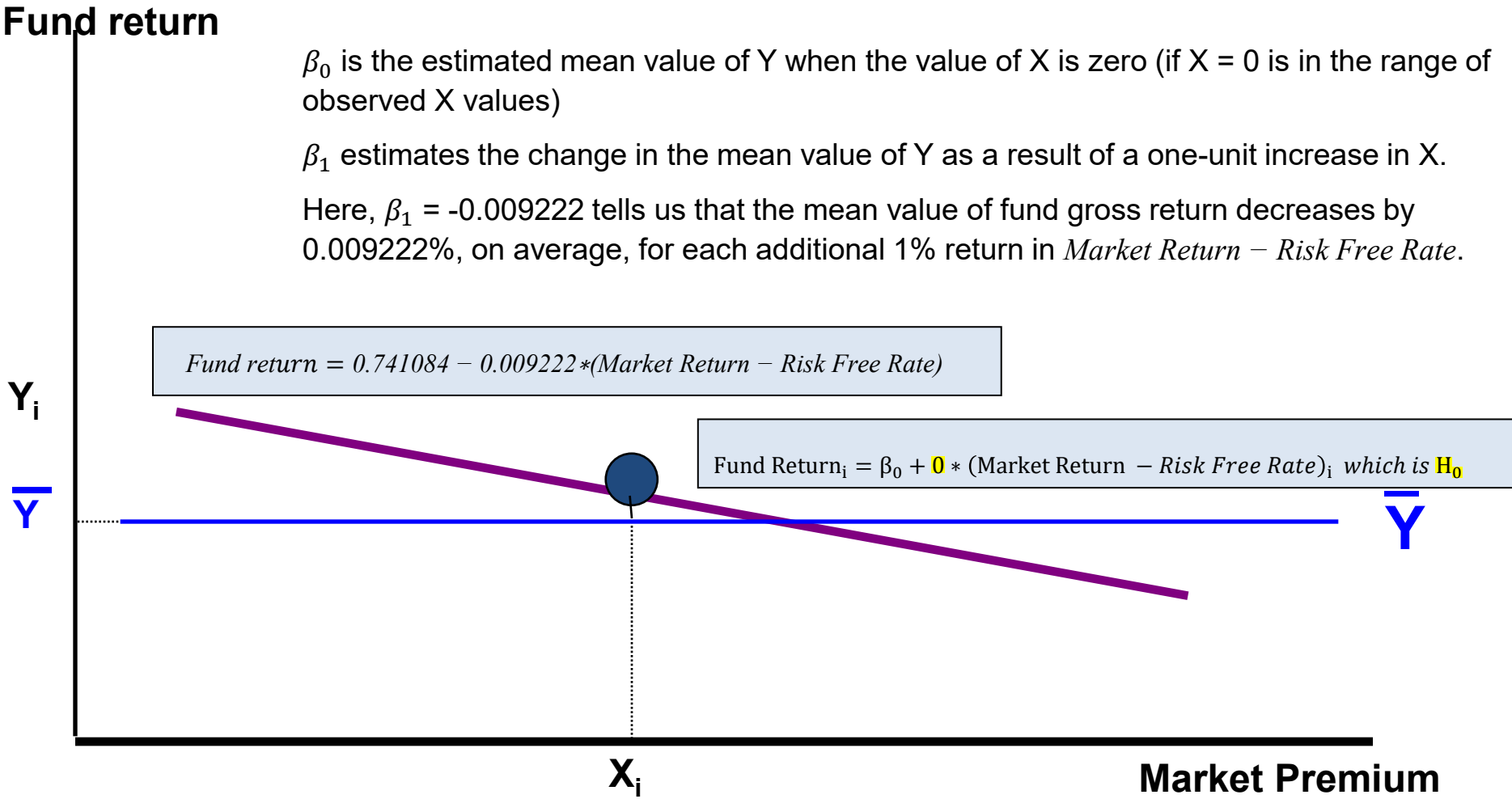
- What data do we need to collect?
- What is the null hypothesis?
- What is the possible outcomes?

Simple Linear Regression

β_0 is the estimated mean value of Y when the value of X is zero (if X = 0 is in the range of observed X values)

β_1 estimates the change in the mean value of Y as a result of a one-unit increase in X.

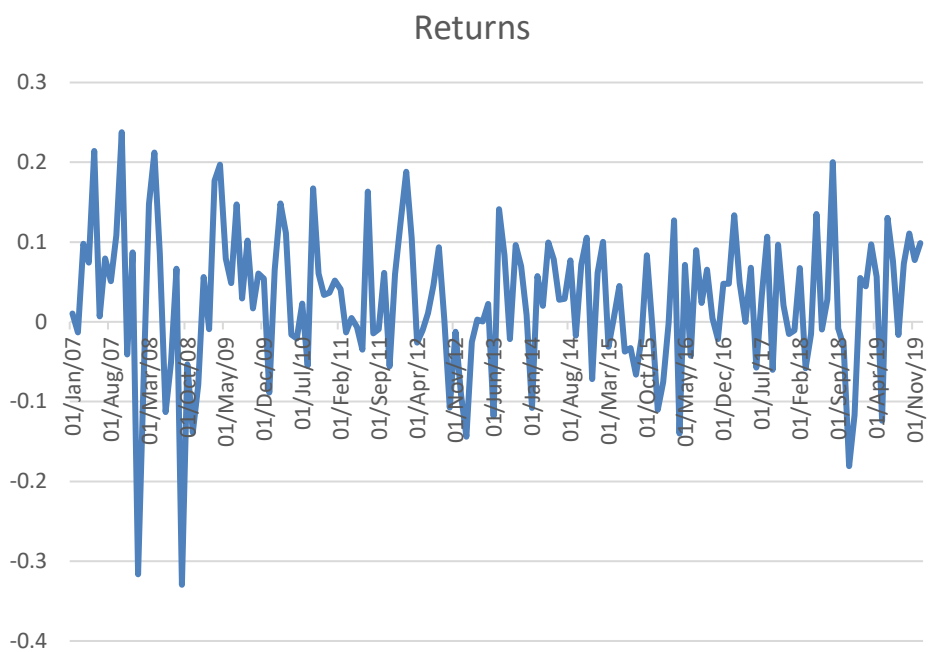
Here, β_1 = -0.009222 tells us that the mean value of fund gross return decreases by 0.009222%, on average, for each additional 1% return in $Market\ Return - Risk\ Free\ Rate$.



Time Series Regression

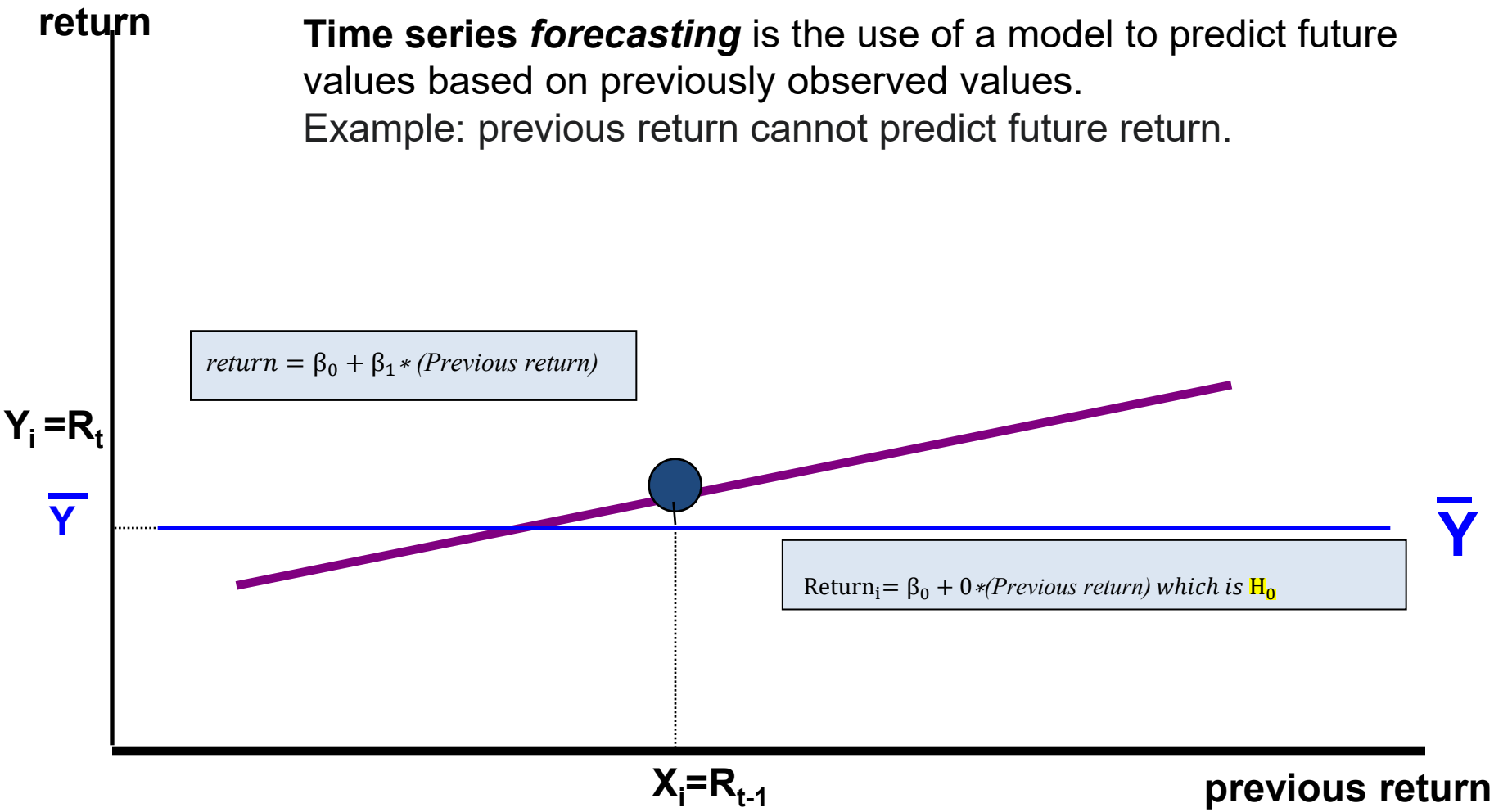
A **time series** is a series of data points indexed (or listed or graphed) in time order. Most commonly, a time series is a sequence taken at successive equally spaced points in time. It is a sequence of discrete-time data. Example: monthly return.

Names Date	Ticker Symbol	Returns
20070131	AAPL	0.01049
20070228	AAPL	-0.013064
20070330	AAPL	0.098097
20070430	AAPL	0.074158
20070531	AAPL	0.214339
20070629	AAPL	0.007005
20070731	AAPL	0.079646
20070831	AAPL	0.051002
20070928	AAPL	0.108247
20071031	AAPL	0.237701
20071130	AAPL	-0.040695
20071231	AAPL	0.087038
20080131	AAPL	-0.31664
20080229	AAPL	-0.076389
20080331	AAPL	0.147816
20080430	AAPL	0.212195
20080530	AAPL	0.085082
20080630	AAPL	-0.112901
20080731	AAPL	-0.050705
20080829	AAPL	0.066562
20080930	AAPL	-0.329558
20081031	AAPL	-0.053405
20081128	AAPL	-0.138675
20081231	AAPL	-0.07899



Time Series Regression

Time series *forecasting* is the use of a model to predict future values based on previously observed values.
Example: previous return cannot predict future return.



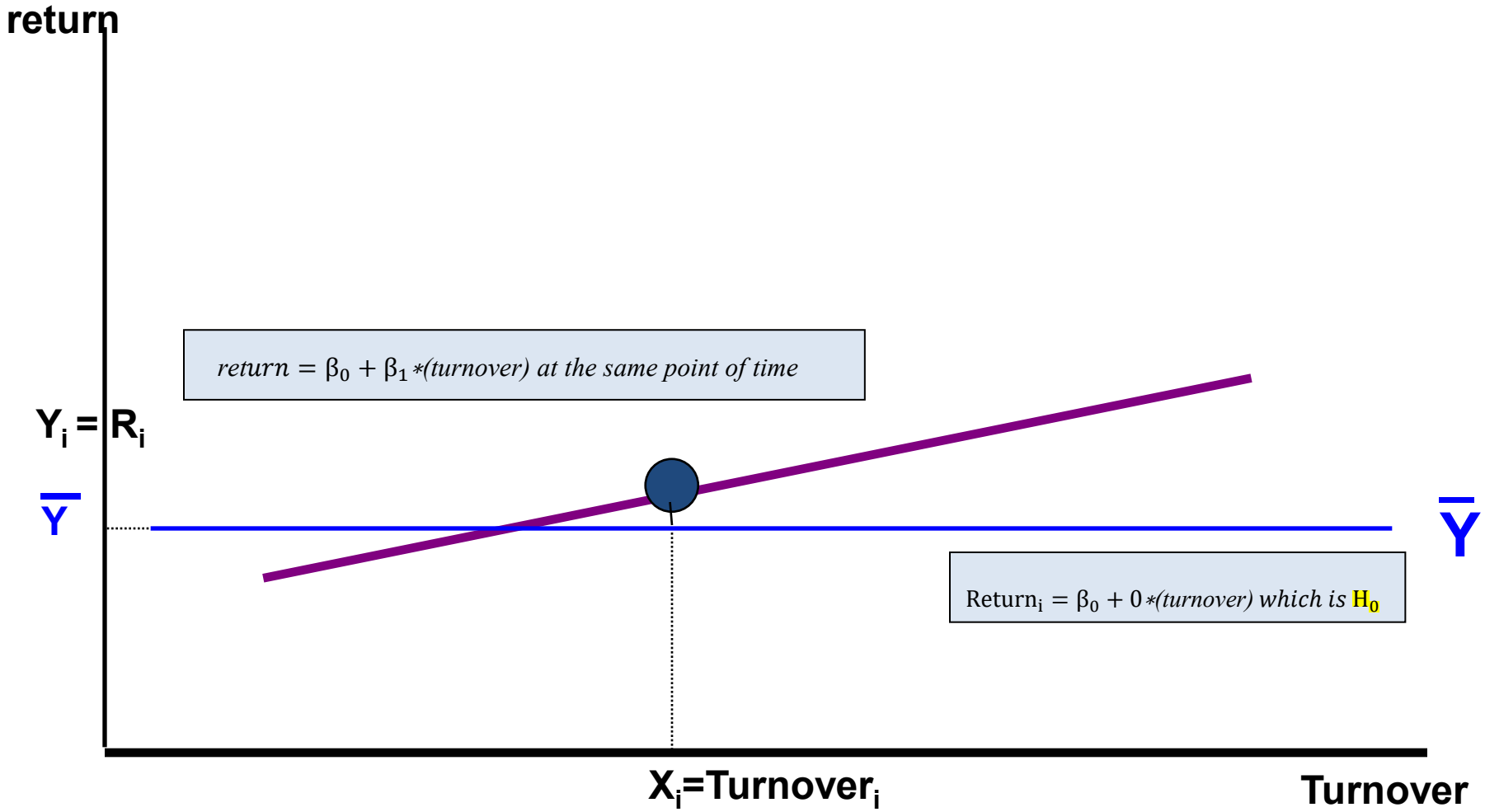
Cross-sectional Regression

Cross-sectional data, or a cross section of a study population, in statistics and econometrics is a type of data collected by observing many subjects (such as individuals, firms, countries, or regions) at the one point or period of time. The analysis might also have no regard to differences in time. Analysis of cross-sectional data usually consists of comparing the differences among selected subjects.

Example: returns of different stocks in 2019

Names Date	Ticker Symbol	Returns	Bid	Ask
20191231	ORCL	-0.056288	52.97	52.98
20191231	MSFT	0.041749	157.48	157.52
20191231	KO	0.036517	55.33	55.34
20191231	XOM	0.024218	69.75	69.77
20191231	FB	0.017903	205.08	205.13
20191231	ABBV	0.009233	88.5	88.52
20191231	PEP	0.013215	136.62	136.66
20191231	CVX	0.028857	120.45	120.48
20191231	AAPL	0.098784	293.38	293.4
20191231	BABA	0.0605	212.11	212.13
20191231	PYPL	0.001481	108.16	108.22
20191231	PDD	0.052017	37.82	37.84
20191231	PG	0.023267	124.82	124.85
20191231	ABT	0.016501	86.84	86.86
20191231	PFE	0.017134	39.16	39.17
20191231	JNJ	0.06095	145.78	145.81
20191231	MRK	0.050241	90.92	90.94
20191231	DIS	-0.040045	144.55	144.58
20191231	JPM	0.057984	139.37	139.4
20191231	DHR	0.052541	153.48	153.49
20191231	WMT	0.002351	118.79	118.82

Cross-sectional Regression



Dummy Variable

Variable A=1 if the event A is true, and zero otherwise
Example: Outperformer =1 if the Returns>Market return, and zero otherwise

Names Date	Ticker Symbol	Returns	Return on the S&P 500 Index	Outperformer
20191231	ORCL	-0.056288	0.02859	0
20191231	MSFT	0.041749	0.02859	1
20191231	KO	0.036517	0.02859	1
20191231	XOM	0.024218	0.02859	0
20191231	FB	0.017903	0.02859	0
20191231	ABBV	0.009233	0.02859	0
20191231	PEP	0.013215	0.02859	0
20191231	CVX	0.028857	0.02859	1
20191231	AAPL	0.098784	0.02859	1
20191231	BABA	0.0605	0.02859	1
20191231	PYPL	0.001481	0.02859	0
20191231	PDD	0.052017	0.02859	1
20191231	PG	0.023267	0.02859	0
20191231	ABT	0.016501	0.02859	0
20191231	PFE	0.017134	0.02859	0
20191231	JNJ	0.06095	0.02859	1
20191231	MRK	0.050241	0.02859	1
20191231	DIS	-0.040045	0.02859	0
20191231	JPM	0.057984	0.02859	1

Does outperformer see higher trading volume?

Dummy Variable

Variable A=1 if the event A is true, and zero otherwise
Example: Outperformer =1 if the Returns>Market return, and zero otherwise

Input

volume c outperformer

Equation: UNTITLED Workfile: TOP50 DATASET::Unti...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: VOLUME
Method: Least Squares
Date: 01/12/21 Time: 00:31
Sample: 1 49
Included observations: 49

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1437833.	360550.0	3.987888	0.0002
OUTPERFORMER	1123488.	515178.8	2.180773	0.0342

R-squared	0.091889	Mean dependent var	1988113.
Adjusted R-squared	0.072567	S.D. dependent var	1871950.
S.E. of regression	1802750.	Akaike info criterion	31.68748
Sum squared resid	1.53E+14	Schwarz criterion	31.76470
Log likelihood	-774.3434	Hannan-Quinn criter.	31.71678
F-statistic	4.755770	Durbin-Watson stat	2.016367
Prob(F-statistic)	0.034240		

Does outperformer see higher trading volume?

$$volume = \beta_0 + \beta_1 * Outperformer$$

Panel Data

Date	Ticker Symbol	Bid or Low Price	Ask or High Price	Price	Volume	Returns	Bid	Ask	Shares Outstanding
20121231	MSFT	26.37	27.68	26.7097	9389131	0.003558	26.72	26.73	8374000
20131231	MSFT	36.25	38.94	37.41	8202590	-0.01888	37.42	37.43	8300000
20141231	MSFT	45.16	48.84	46.45	6287260	-0.02845	46.46	46.47	8218000
20151231	MSFT	54.06	56.55	55.48	7934234	0.020791	55.45	55.46	7925000
20161230	MSFT	59.2	63.62	62.14	5135897	0.031198	62.15	62.16	7730000
20171229	MSFT	81.08	86.85	85.54	4663253	0.016277	85.59	85.6	7705000
20181231	MSFT	94.13	112.09	101.57	9443269	-0.08405	101.66	101.67	7683000
20191231	MSFT	149.31	158.96	157.7	4520222	0.041749	157.48	157.52	7632139
20121231	AAPL	509.5892	586.19	532.1729	4472556	-0.09074	533.02	533.16	938973
20131231	AAPL	544.46	570.09	561.02	2547642	0.008902	560.82	561.16	892447
20141231	AAPL	106.745	115.93	110.38	10766123	-0.07189	110.38	110.4	5826419
20151231	AAPL	105.26	119.03	105.26	9237703	-0.11023	105.22	105.23	5544487
20161230	AAPL	109.11	117.26	115.82	6107580	0.047955	115.84	115.87	5257816
20171229	AAPL	169.01	176.42	169.23	5319099	-0.01525	169.29	169.3	5087056
20181231	AAPL	146.83	184.82	157.74	8989251	-0.1167	157.93	157.94	4729803
20191231	AAPL	259.45	293.65	293.65	5984993	0.098784	293.38	293.4	4443265
20141231	BABA	103.88	110.65	103.94	3555254	-0.06897	103.94	103.95	498068
20151231	BABA	79.74	85	81.27	3139190	-0.03342	81.25	81.27	997391
20161230	BABA	86.79	93.28	87.81	2181329	-0.06605	87.79	87.8	1061184
20171229	BABA	168.96	179.29	172.43	4162459	-0.02626	172.42	172.43	1135511
20181231	BABA	131.89	163.74	137.07	3445628	-0.14789	137.06	137.07	1135511
20191231	BABA	193.74	216.38	212.1	3012111	0.0605	212.11	212.13	1135511
20121231	GOOG	684.21	722.36	707.38	444030	0.012901	709.26	709.37	267448
20131231	GOOG	1053.26	1120.71	1120.71	343998	0.057683	1119.96	1121	279325

Panel Data

Please identify the dummy variables in the following tables

Bernile, G., Bhagwat, V. and Rau, P.R., 2017. **What doesn't kill you will only make you more risk-loving: Early-life disasters and CEO behavior**. *The Journal of Finance*, *72*(1), pp.167-206.

Panel C: Top 10 CEO Birth States and Disaster Experience

Top 10 Birth States for All CEOs	No. of CEOs	%No Fatality CEOs	%Medium Fatality CEOs	%Extreme Fatality CEOs
New York	251	27.49%	61.35%	11.16%
Illinois	116	33.62%	55.17%	11.21%
Pennsylvania	110	20.91%	59.09%	20.00%
Ohio	84	26.19%	65.48%	8.33%
California	71	25.35%	64.79%	9.86%
Massachusetts	69	30.43%	57.97%	11.59%
New Jersey	64	37.50%	53.13%	9.38%
Texas	60	38.33%	55.00%	6.67%
Missouri	43	30.23%	48.84%	20.93%
Iowa	41	17.07%	68.29%	14.63%

Panel D: Top 10 CEO Birth States by Disaster Experience Categories

No Fatality CEOs		Medium Fatality CEOs		Extreme Fatality CEOs	
Top 10 Birth States	No. of CEOs	Top 10 Birth States	No. of CEOs	Top 10 Birth States	No. of CEOs
New York	69	New York	154	New York	28
Illinois	39	Pennsylvania	65	Pennsylvania	22
New Jersey	24	Illinois	64	Illinois	13
Pennsylvania	23	Ohio	55	Missouri	9
Texas	23	California	46	Massachusetts	8
Ohio	22	Massachusetts	40	California	7
Massachusetts	21	New Jersey	34	Ohio	7
California	18	Texas	33	Kentucky	7
Missouri	13	Iowa	28	Washington	6
Mississippi	12	Indiana	28	New Jersey	6

The modal birth state is New York, which accounts for 251 CEOs in our sample. Of these,27.49% did not experience any fatal disasters in their birth county between the ages of 5 and 15, 11.16% experienced extreme levels of fatal disasters, while the remaining 61.35% fall in to the Medium Fatality group.

Panel Regression

CEO Early-Life Disaster Experience and Corporate Financial Policies

	Book Leverage			Cash-to-Asset		
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Medium Fatality Experience	0.033***	0.030***	0.030***	–0.012**	–0.010**	–0.010**
	(0.009)	(0.009)	(0.008)	(0.005)	(0.005)	(0.005)
Extreme Fatality Experience	–0.035***	–0.032***	–0.035***	0.022***	0.022***	0.020***
	(0.011)	(0.011)	(0.011)	(0.007)	(0.007)	(0.007)
Average Fatality Risk of County	0.003			–0.005		
	(0.007)			(0.004)		
(Average Fatality Risk of County)^2	–0.000			0.001		
	(0.001)			(0.001)		
Ln(Book Assets)	–0.010			–0.030***		
	(0.009)			(0.006)		
Market-to-Book	–0.011***			0.005*		
	(0.002)			(0.002)		
Fixed Assets/Book Assets	0.040			–0.294***		
	(0.061)			(0.034)		
Dividend Paying	–0.023**			–0.001		
	(0.012)			(0.006)		
ROA	–0.116***			0.032		
	(0.026)			(0.022)		
Sales Growth	0.001			–0.005**		
	(0.003)			(0.002)		
CEO Age	–0.000			–0.006		
	(0.007)			(0.005)		
CEO Age^2	0.000			0.000*		
	(0.000)			(0.000)		
CEO is Female	0.021			0.008		
	(0.019)			(0.012)		
Observations	8,533	8,533	8,533	8,533	8,533	8,533
Adjusted R^2	0.828	0.799	0.810	0.842	0.822	0.837
3rd-Order Polynomial Controls	No	Yes	No	No	Yes	No
Decile Step Function Controls	No	No	Yes	No	No	Yes
Firm, Year, Birth Year, and State of Birth FE	Yes	Yes	Yes	Yes	Yes	Yes

Column 1: All else equal, firms whose CEOs experienced a moderate level of fatalities from natural disasters have 3.3% *higher* leverage than firms whose CEOs have no fatal disaster experience. In contrast, firms whose CEOs experienced an extreme level of fatalities from disasters have 3.5% *lower* leverage than firms whose CEOs have no fatal disaster experience

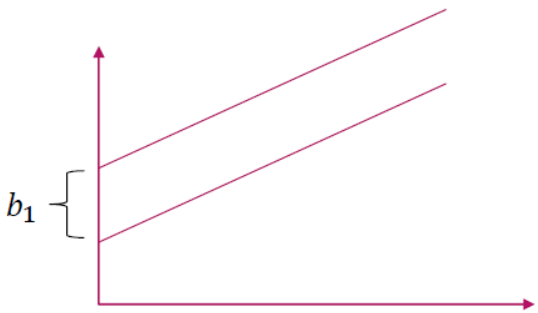
Column 4: All else equal, firms whose CEOs experienced a moderate level of fatalities from natural disasters hold 1.2% *less* cash per dollar of assets than firms whose CEOs experienced no fatal disasters. In contrast, firms whose CEOs experienced an extreme level of fatalities from disasters hold 2.2% *more* cash per dollar of assets than firms whose CEOs have no fatal disaster experience.

Dummy Variable and Event Study

$y = a_0 + b_1 D + b_2 X + e$

When $D=0$: $y = a_0 + b_2 X + e$

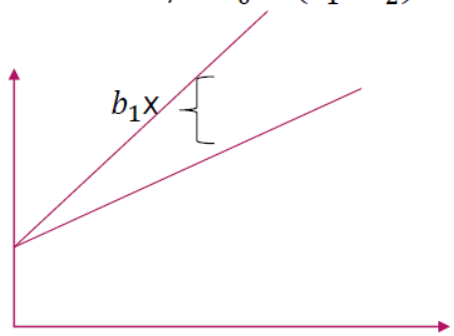
When $D=1$: $y = a_0 + b_1 + b_2 X + e$



$y = a_0 + b_1 (D * X) + b_2 X + e$

When $D=0$: $y = a_0 + b_2 X + e$

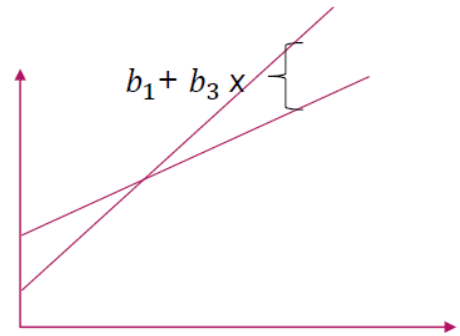
When $D=1$: $y = a_0 + (b_1 + b_2) X + e$



$y = a_0 + b_1 D + b_2 X + b_3 (D * X) + e$

When $D=0$: $y = a_0 + b_2 X + e$

When $D=1$: $y = a_0 + b_1 + (b_3 + b_2) X + e$



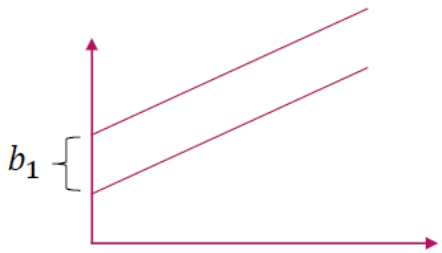
Dummy Variable and Event Study

Dummy variable: D=1 if event happens and D=0 if event does not happen

$$y = a_0 + b_1 D + b_2 X + e$$

When $D=0$: $y = a_0 + b_2 X + e$

When $D=1$: $y = a_0 + b_1 + b_2 X + e$



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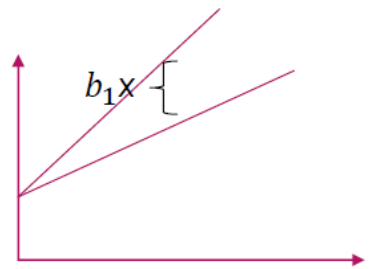
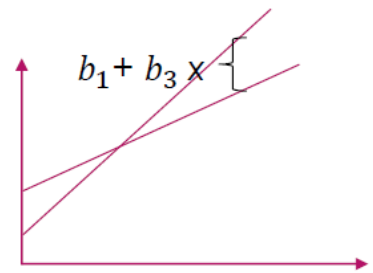




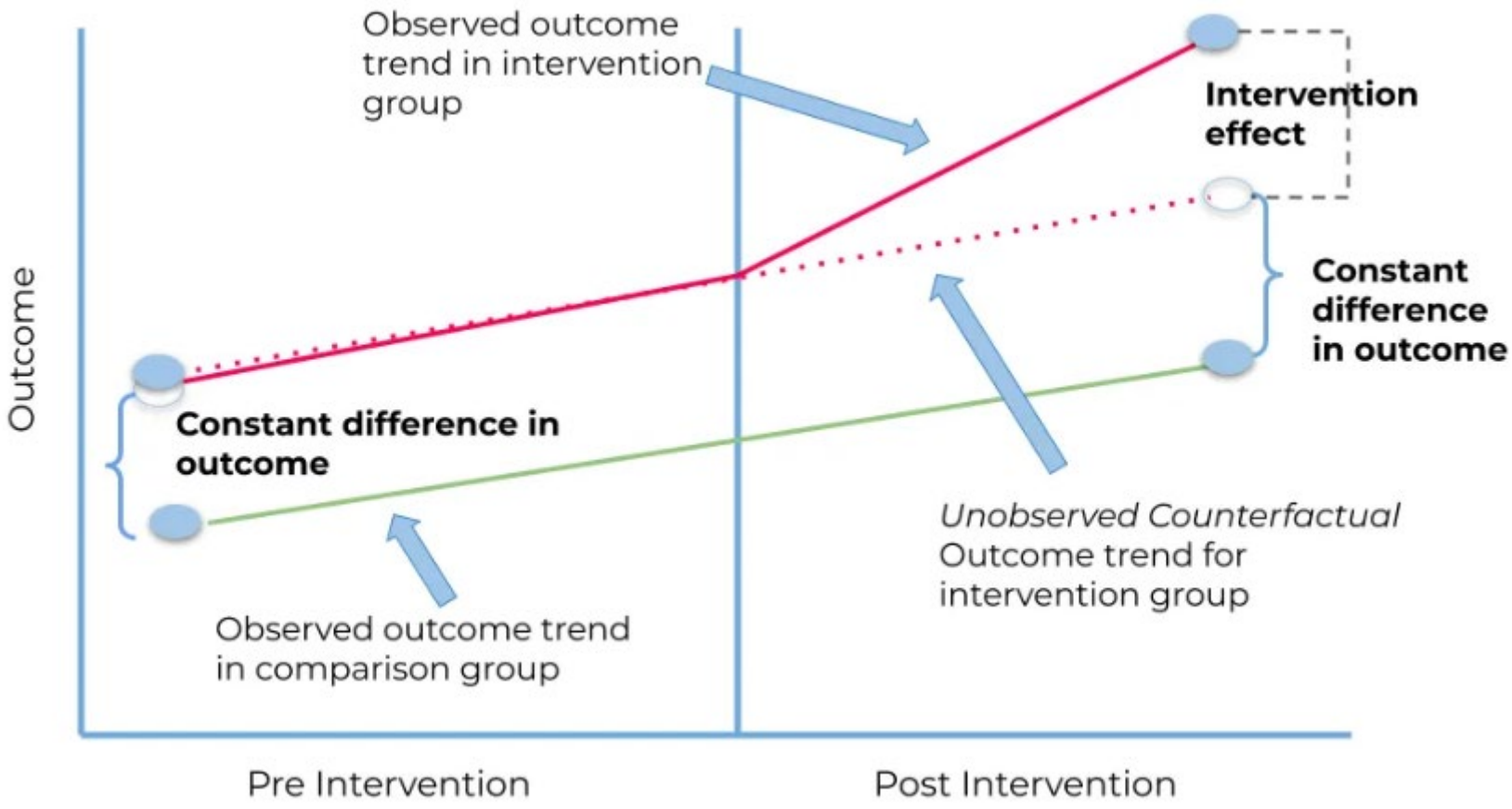
E.g. D=1 if the company belongs to financial industry;
D=1 if the board of the company has female;
D=1 if there is stock dividend;
D=1 if the firm's ESG score is above industry average
D=1 if the variable is observed during COVID-19

$$y = a_0 + b_1 D + b_2 X + b_3 (D * X) + e$$

When $D=0$: $y = a_0 + b_2 X + e$

When $D=1$: $y = a_0 + b_1 + (b_3 + b_2) X + e$

Difference-in-difference (DiD)



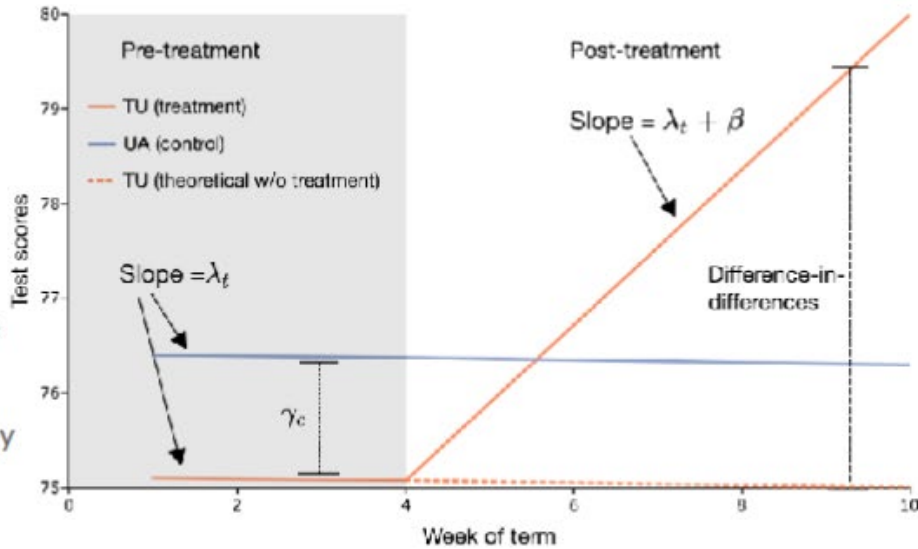
Difference-in-difference (DiD)

$$y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \cdot S) + \varepsilon$$

y_{st}	$s = 2$	$s = 1$	**Difference**
$t = 2$	y_{22}	y_{12}	$y_{12} - y_{22}$
$t = 1$	y_{21}	y_{11}	$y_{11} - y_{21}$
Change	$y_{21} - y_{22}$	$y_{11} - y_{12}$	$(y_{11} - y_{21}) - (y_{12} - y_{22})$

Key assumptions (Angrist and Pischke, 2008):

- Outcomes in the treatment group and the control group follow the same trend, λ_t
- The treatment causes deviation, β, from the trend.
- The differences in the treatment group and control group are captured by the fixed effects variables, γ_c.



Difference-in-difference (DiD)

$$y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \cdot S) + \varepsilon$$

y_{st}	$s = 2$	$s = 1$	Difference
$t = 2$	y_{22}	y_{12}	$y_{12} - y_{22}$
$t = 1$	y_{21}	y_{11}	$y_{11} - y_{21}$
Change	$y_{21} - y_{22}$	$y_{11} - y_{12}$	$(y_{11} - y_{21}) - (y_{12} - y_{22})$

where T is a dummy variable for the period, equal to 1 when t=2, and S is a dummy variable for group membership, equal to 1 when s=2. The composite variable (T·S) is a dummy variable indicating when T=S=1.

$$\hat{\beta}_0 = \widehat{E}(y \mid T = 0,\ S = 0)$$

$$\hat{\beta}_1 = \widehat{E}(y \mid T = 1,\ S = 0) \quad \widehat{E}(y \mid T = 0,\ S = 0)$$

$$\hat{\beta}_2 = \widehat{E}(y \mid T = 0,\ S = 1) - \widehat{E}(y \mid T = 0,\ S = 0)$$

$$\hat{\beta}_3 = \left[\widehat{E}(y \mid T = 1,\ S = 1) - \widehat{E}(y \mid T = 0,\ S = 1)\right] - \left[\widehat{E}(y \mid T = 1,\ S = 0) - \widehat{E}(y \mid T = 0,\ S = 0)\right],$$

$$\hat{\beta}_3 = (y_{11} - y_{21}) - (y_{12} - y_{22}).$$

Difference-in-difference (DiD)

$$y = \beta_0 + \beta_1 T + \beta_2 S + \beta_3 (T \cdot S) + \varepsilon$$

	New Jersey	Pennsylvania	Difference
February	20.44	23.33	−2.89
November	21.03	21.17	−0.14
Change	0.59	−2.16	2.75

Card and Krueger (1994) compared employment in the fast food sector in New Jersey and in Pennsylvania, in February 1992 and in November 1992, after New Jersey's minimum wage rose from $4.25 to $5.05 in April 1992. Observing a change in employment in New Jersey only, before and after the treatment, would fail to control for omitted variables such as weather and macroeconomic conditions of the region.

New Jersey and Pennsylvania have parallel trends over time, Pennsylvania's change in employment can be interpreted as the change New Jersey would have experienced, had they not increased the minimum wage, and vice versa.

Card and Krueger estimate that the $0.80 minimum wage increase in New Jersey led to a 2.75 full-time equivalents increase in employment.

Panel Data regression

Panel data models provide information on individual behavior, both across individuals and over time. The data and models have both cross-sectional and time-series dimensions. Panel data can be balanced when all individuals are observed in all time periods or unbalanced when individuals are not observed in all time periods. Examples include estimating the effect of education on income, with data across time and individuals

Panel data regression model:

$$Y_i = \beta_0 + \beta_1 X_{it} + \beta_2 Z_i + u_{it}$$

where the Zi are unobserved time-invariant heterogeneities across the assets i=1,…,n.

Fixed Effects

Fixed effects are variables that are constant across individuals; these variables, like age, sex, or ethnicity, don't change or change at a constant rate over time. They have **fixed effects**; in other words, any change they cause to an individual is the same.

Panel data regression model:

$$Y_i = \beta_0 + \beta_1 X_{it} + \beta_2 Z_i + u_{it}$$

where the Zi are unobserved time-invariant heterogeneities across the assets i=1,…,n. Letting $\alpha_i = \beta_0 + \beta_2 Z_i$ we have $Y_{it} = \alpha_i + \beta_1 X_{it} + u_{it}$. Vith individual specific intercepts for i=1,…,n, each of these can be understood as the fixed effect for asset i. this is equivalent to a regression with n-1 dummy regressors and a constant:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i + \gamma_3 D3_i + \cdots + \gamma_n Dn_i + u_{it}.$$

Year Fixed Effect

City and year dummy: $murder_{it} = \beta_0 + \beta_1 popden_{it} + \alpha_2 City2 - \alpha_3 City3 + \delta_2 Yr2001 + \delta_3 Yr2002 + u_{it}$

obs	i	t	murder rate	pop density	City1	City2	City3	Yr00	Yr01	Yr02
1	1	2000	9.3	2.24	1	0	0	1	0	0
2	1	2001	11.6	2.38	1	0	0	0	1	0
3	1	2002	11.8	2.42	1	0	0	0	0	1
4	2	2000	7.6	1.61	0	1	0	1	0	0
5	2	2001	10.3	1.73	0	1	0	0	1	0
6	2	2002	11.9	1.81	0	1	0	0	0	1
7	3	2000	11.1	6.00	0	0	1	1	0	0
8	3	2001	17.2	6.33	0	0	1	0	1	0
9	3	2002	20.3	6.42	0	0	1	0	0	1

When city3=1 (city2=0):

$$murders_{3t} = \beta_0 + \beta_1 popden_{3t} + \alpha_2 \cdot 0 + \alpha_3 \cdot 1 + \delta_2 Yr2001 + \delta_3 Yr2002 + u_{3t}$$

$$murders_{3t} = \beta_0 + \beta_1 popden_{3t} + \alpha_3 + \delta_2 Yr2001 + \delta_3 Yr2002 + u_{3t}$$

City3 fixed effect

- Follow the literature to control fixed effects such as cities, counties, states, countries or even units like individuals, households, etc.
- Likewise the time units might be days, months, years, etc.

```
2015   921299 108 0   7.543635   4.595307 1 0 1.3862944   2.772589    .10401749 0 0 1
```

Year Fixed Effects

Source	SS	df	MS
Model	17587.3082	17	1034.54754
Residual	61489.7938	1,198	51.3270399
Total	79077.102	1,215	65.0840345

Number of obs	=	1,216
F(17, 1198)	=	20.16
Prob > F	=	0.0000
R-squared	=	0.2224
Adj R-squared	=	0.2114
Root MSE	=	7.1643

bogindex_w	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
egc	1.455403	.6179286	2.36	0.019	.2430606	2.667746
logSIZE_w	.0537797	.2313815	0.23	0.816	-.4001784	.5077378
MTB_w	-.3560624	.1056944	-3.37	0.001	-.5634291	-.1486957
AGE_IPO	.6235395	.2403697	2.59	0.010	.151947	1.095132
spiDummy	-3.078925	.4488517	-6.86	0.000	-3.959548	-2.198302
LOGoneplusNBSEG_w	-.8499284	.5376904	-1.58	0.114	-1.904848	.2049912
LOGoneplusNGSEG_w	-3.351886	.3561815	-9.41	0.000	-4.050695	-2.653077
ROA_w	8.455898	2.777385	3.04	0.002	3.006819	13.90498
LogNITEMS_w	-3.303121	4.2201	-0.78	0.434	-11.58273	4.976487
MAdummy	-1.995824	1.146018	-1.74	0.082	-4.24425	.2526022
SEOdummy	1.980199	.4816958	4.11	0.000	1.035138	2.925261
DummyDLW	.6609418	1.343901	0.49	0.623	-1.97572	3.297604
fyear						
2013	5.618195	2.044632	2.75	0.006	1.606738	9.629652
2014	7.249985	1.967891	3.68	0.000	3.389088	11.11088
2015	8.278689	1.972914	4.20	0.000	4.407937	12.14944
2016	8.499862	1.990884	4.27	0.000	4.593855	12.40587
2017	8.921632	2.011748	4.43	0.000	4.974691	12.86857
_cons	107.2165	23.36503	4.59	0.000	61.37559	153.0574

Year Fixed Effect

City and year dummy: $murder_{it} = \beta_0 + \beta_1 popden_{it} + \alpha_2 City2 - \alpha_3 City3 + \delta_2 Yr2001 + \delta_3 Yr2002 + u_{it}$

obs	i	t	murder rate	pop density	City1	City2	City3	Yr00	Yr01	Yr02
1	1	2000	9.3	2.24	1	0	0	1	0	0
2	1	2001	11.6	2.38	1	0	0	0	1	0
3	1	2002	11.8	2.42	1	0	0	0	0	1
4	2	2000	7.6	1.61	0	1	0	1	0	0
5	2	2001	10.3	1.73	0	1	0	0	1	0
6	2	2002	11.9	1.81	0	1	0	0	0	1
7	3	2000	11.1	6.00	0	0	1	1	0	0
8	3	2001	17.2	6.33	0	0	1	0	1	0
9	3	2002	20.3	6.42	0	0	1	0	0	1

When city3=1 (city2=0): $murders_{3t} = \beta_0 + \beta_1 popden_{3t} + \alpha_2 \cdot 0 + \alpha_3 \cdot 1 + \delta_2 Yr2001 + \delta_3 Yr2002 + u_{3t}$

$murders_{3t} = \beta_0 + \beta_1 popden_{3t} + \alpha_3 + \delta_2 Yr2001 + \delta_3 Yr2002 + u_{3t}$

City3 fixed effect

- Follow the literature to control fixed effects such as cities, counties, states, countries or even units like individuals, households, etc.
- Likewise the time units might be days, months, years, etc.

Fixed Effects

CEO Early-Life Disaster Experience and Corporate Financial Policies

	Book Leverage			Cash-to-Asset		
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Medium Fatality Experience	0.033***	0.030***	0.030***	–0.012**	–0.010**	–0.010**
	(0.009)	(0.009)	(0.008)	(0.005)	(0.005)	(0.005)
Extreme Fatality Experience	–0.035***	–0.032***	–0.035***	0.022***	0.022***	0.020***
	(0.011)	(0.011)	(0.011)	(0.007)	(0.007)	(0.007)
Average Fatality Risk of County	0.003			–0.005		
	(0.007)			(0.004)		
(Average Fatality Risk of County)^2	–0.000			0.001		
	(0.001)			(0.001)		
Ln(Book Assets)	–0.010			–0.030***		
	(0.009)			(0.006)		
Market-to-Book	–0.011***			0.005*		
	(0.002)			(0.002)		
Fixed Assets/Book Assets	0.040			–0.294***		
	(0.061)			(0.034)		
Dividend Paying	–0.023**			–0.001		
	(0.012)			(0.006)		
ROA	–0.116***			0.032		
	(0.026)			(0.022)		
Sales Growth	0.001			–0.005**		
	(0.003)			(0.002)		
CEO Age	–0.000			–0.006		
	(0.007)			(0.005)		
CEO Age^2	0.000			0.000*		
	(0.000)			(0.000)		
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	(0.019)			(0.012)		
Observations	8,533	8,533	8,533	8,533	8,533	8,533
Adjusted R^2	0.828	0.799	0.810	0.842	0.822	0.837
3rd-Order Polynomial Controls	No	Yes	No	No	Yes	No
Decile Step Function Controls	No	No	Yes	No	No	Yes
Firm, Year, Birth Year, and State of Birth FE	Yes	Yes	Yes	Yes	Yes	Yes

For a panel data with 1000 firms over 20 years, how to control for fixed effects?

Subsample Analysis

Subsampling, i.e. **analyses** of a fraction of the sample and subsequent extrapolation, can be a suitable strategy to reduce the effort of sample **analysis**.

Subsamples

Time-varying Firm Characteristics

	All Observations ($N = 8{,}533$)		No Fatality ($N = 2{,}836$)		Medium Fatality ($N = 4{,}864$)		Extreme Fatality ($N = 833$)	
Sample	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Book Assets ($M)	30,863	119,903	18,148	57,533	36,787	141,925	39,562	132,615
Market-to-Book ratio	1.84	1.41	1.81	1.24	1.86	1.52	1.88	1.30
Book Leverage	0.25	0.18	0.26	0.18	0.26	0.19	0.24	0.18
Cash-to-Assets	0.11	0.14	0.11	0.14	0.11	0.14	0.13	0.17
I(Dividend Paying)	0.73	0.45	0.73	0.44	0.72	0.45	0.76	0.43
ROA	0.05	0.13	0.05	0.11	0.05	0.13	0.06	0.19
Stock Volatility (%)	38.60	22.09	38.44	20.83	39.01	23.15	36.81	19.67
Idiosyncratic Volatility (%)	34.47	20.72	34.67	19.96	34.76	21.52	32.11	18.26
CEO Age	56.85	7.01	56.96	6.91	56.83	7.15	56.59	6.49
I(Announced Acquisition)	0.42	0.49	0.42	0.49	0.43	0.50	0.33	0.47
I(Bankrupt Next Year)	0.01	0.07	0.01	0.06	0.01	0.08	<0.01	0.03
Interest Expense/Debt	0.08	0.07	0.08	0.06	0.08	0.07	0.07	0.06

Subsample Analysis

- Simple linear regression - Hirshleifer, D. and Shumway, T., 2003. Good day sunshine: Stock returns and the weather. The Journal of Finance, 58(3), pp.1009-1032.
  - Explain returns (r) using the weather (SKC the larger the cloudier) city-by-city

	OLS Regression		
Location	Observations (2)	β_{iC} (3)	t-Statistic (4)
Amsterdam	3984	−0.007	−1.07
Athens	2436	0.012	0.71
Bangkok	3617	0.009	0.45
Brussels	3997	−0.018*	−3.25
Buenos Aires	2565	−0.030	−0.98
Copenhagen	4042	−0.002	0.30
Dublin	3963	−0.000	−0.02
Helsinki	2725	−0.016	−1.67
Istanbul	2500	0.007	0.32
Johannesburg	3999	0.004	0.47
Kuala Lumpur	3863	0.014	0.26
London	4003	0.010	−1.52
Madrid	3760	−0.011	−1.60
Manila	2878	0.018	0.83
Milan	3961	−0.014*	2.03
New York	4013	−0.007	−1.28
Oslo	3877	−0.018	−1.92
Paris	3879	−0.009	−1.27
Rio de Janeiro	2988	−0.057	−1.93
Santiago	2636	0.000	0.05
Singapore	3890	0.008	0.37
Stockholm	3653	−0.011	−1.54
Sydney	4037	0.014*	−1.96
Taipei	3784	0.016	−0.97
Vienna	3907	−0.013*	2.14
Zurich	3851	−0.007	−1.28
All cities (naive)	92808	0.011*	−4.49
All cities (PCSE)	92808	−0.010*	3.96

$$r_{it} = \alpha_t + \beta_{iC}\mathrm{SKC}^*_{it} + e_{it}$$

Four of the negative estimated coefficients are significantly different from zero using a two-tailed, five percent test.

The coefficient on SKC, reported in column 3, is positive for 8 out of 26 cities. If the sign of each regression is an independent draw from the binomial distribution, and if the probability of drawing a negative coefficient is 0.5, then the probability of finding only 8 positive coefficients out of 26 possible is 0.038. This is within the five percent level of significance for a one tailed test. The simple regression coefficients reported in Table III suggest that cloudiness and returns are correlated.

Logit Regression

The population Logit regression function is

$$P(Y=1|X_1,X_2,\ldots,X_k) = F(\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_kX_k)$$
$$= \frac{1}{1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_kX_k)}}.$$

The idea is similar to Probit regression except that a different CDF is used:

$$F(x) = \frac{1}{1+e^{-x}}$$

is the CDF of a standard logistically distributed random variable.

Logit Regression

- Logit regression - Campbell, J.Y., Hilscher, J. and Szilagyi, J., 2008. In search of distress risk. *The Journal of Finance*, *63*(6), pp.2899-2939.
  - Explain probability(bankruptcy or failures) using financial ratios $P_{i,t-1}(Y_{it}=1)=\frac{1}{1+\exp(-\alpha-\beta x_{i,t-1})}$

	Model 1			Model 2		
Dependent variable:	Bankruptcy	Failure	Failure	Bankruptcy	Failure	Failure
Sample period:	1963–1998	1963–1998	1963–2003	1963–1998	1963–1998	1963–2003
NITA	−14.05	−13.79	−12.78			
	(16.03)**	(17.06)**	(21.26)**			
NIMTAAVG				−32.52	−32.46	−29.67
				(17.65)**	(19.01)**	(23.37)**
TLTA	5.38	4.62	3.74			
	(25.91)**	(26.28)**	(32.32)**			
TLMTA				4.32	3.87	3.36
				(22.82)**	(23.39)**	(27.80)**
EXRET	3.30	2.90	3.32			
	(12.12)**	(11.81)**	(13.57)**			
EXRETAVG				−9.51	−9.82	−7.35
				(12.05)**	(12.08)**	(14.03)**
SIGMA	2.15	2.28	2.76	0.920	1.15	1.48
	(16.40)**	(18.34)**	(26.62)**	(6.66)**	(8.79)**	(13.54)**
RSIZE	−0.188	−0.253	−0.374	0.246	0.169	0.082
	(5.56)**	(7.60)**	(13.28)**	(6.18)**	(4.32)**	(2.62)**
CASHMTA				−4.89	−3.22	−2.40
				(7.88)**	(6.59)**	(8.64)**	
MB				0.099	0.095	0.054
				(6.72)**	(6.76)**	(4.87)**
PRICE				0.882	0.807	0.937
				(10.39)**	(10.09)**	(14.77)**
Constant	15.21	15.41	16.58	7.65	8.45	9.08
	(39.45)**	(40.87)**	(50.92)**	(13.66)**	(15.63)**	(20.84)**
Observations	1,282,853	1,302,564	1,695,036	1,282,853	1,302,564	1,695,036
Failures	797	911	1,614	797	911	1,614
Pseudo-R^2	0.260	0.258	0.270	0.299	0.296	0.312

To get some idea of the relative impact of changes in the different variables, we compute the proportional impact on the failure probability of a one-standard deviation increase in each predictor variable for a firm that initially has sample mean values of the predictor variables. Such an increase in profitability reduces the probability of failure by 44% of its initial value; the corresponding effects are a 156% increase for leverage, a 28% reduction for past excess return, a 64% increase for volatility, a 17% increase for market capitalization, a 21% reduction for cash holdings, a 9% increase for the market-to-book ratio, and a 56% reduction for price per share. Thus, variations in leverage, volatility, price per share, and profitability are more important for failure risk than movements in market capitalization, cash, or the market-to-book ratio. These magnitudes roughly line up with the *t*-statistics reported in Table III

Fama-Macbeth (1973) regression

- First regress each of *n* asset returns against *m* proposed risk factors to determine each asset's beta exposures.

$$
\begin{aligned}
R_{1,t} &= \alpha_1 + \beta_{1,F_1} F_{1,t} + \beta_{1,F_2} F_{2,t} - \cdots + \beta_{1,F_m} F_{m,t} + \epsilon_{1,t} \\
R_{2,t} &= \alpha_2 + \beta_{2,F_1} F_{1,t} + \beta_{2,F_2} F_{2,t} - \cdots + \beta_{2,F_m} F_{m,t} + \epsilon_{2,t} \\
&\vdots \\
R_{n,t} &= \alpha_n + \beta_{n,F_1} F_{1,t} + \beta_{n,F_2} F_{2,t} + \cdots + \beta_{n,F_m} F_{m,t} + \epsilon_{n,t}
\end{aligned}
$$

- Second regress all asset returns for each of *T* time periods against the previously estimated betas to determine the risk premium for each factor.

$$
\begin{aligned}
R_{i,1} &= \gamma_{1,0} + \gamma_{1,1}\hat{\beta}_{i,F_1} + \gamma_{1,2}\hat{\beta}_{i,F_2} + \cdots + \gamma_{1,m}\hat{\beta}_{i,F_m} + \epsilon_{i,1} \\
R_{i,2} &= \gamma_{2,0} + \gamma_{2,1}\hat{\beta}_{i,F_1} + \gamma_{2,2}\hat{\beta}_{i,F_2} + \cdots + \gamma_{2,m}\hat{\beta}_{i,F_m} + \epsilon_{i,2} \\
&\vdots \\
R_{i,T} &= \gamma_{T,0} + \gamma_{T,1}\hat{\beta}_{i,F_1} + \gamma_{T,2}\hat{\beta}_{i,F_2} + \cdots + \gamma_{T,m}\hat{\beta}_{i,F_m} + \epsilon_{i,T}
\end{aligned}
$$

Fama-Macbeth (1973) regression

- S. Roychowdhury / Journal of Accounting and Economics 42 (2006) 335–370
  - Comparison of suspect firm-years with the rest of the sample

$$Y_t = \alpha + \beta_1(\text{SIZE})_{t-1} + \beta_2(\text{MTB})_{t-1} + \beta_3(\text{Net income})_t + \beta_4(\text{SUSPECT_NI})_t + \varepsilon_t.$$

Table 4
Comparison of suspect firm-years with the rest of the sample

	Abnormal CFO	Abnormal discretionary expenses	Abnormal production costs
Intercept	0.0026**	0.0464**	−0.0021**
	(7.40)	(11.94)	(−2.51)
SIZE	0.0001	0.0237**	−0.0041**
	(0.56)	(7.34)	(−3.96)
MTB	0.0010**	0.0033**	−0.0039**
	(3.11)	(2.00)	(−6.36)
Net income	0.1904**	−0.1721**	−0.1118**
	(7.21)	(−4.65)	(−6.02)
SUSPECT_NI	0.0200**	0.0591**	0.0497**
	(−3.05)	(−4.35)	(4.99)

The first two columns in Table 4 provide evidence on H1A—both abnormal CFO and abnormal discretionary expenses are unusually low for suspect firm-years. When the dependent variable in regression (6) is abnormal CFO, the coefficient on SUSPECT_NI is negative (0.0200) and significant at the 5% level (t =-3.05). Suspect firm-years have abnormal CFO that is lower on average by 2% of assets compared to the rest of the sample. This difference is economically large, given that the median CFO across all firm-years is 8% of total assets at the beginning of the year (see Table 1).

Fama-Macbeth (1973) regression

- Campbell, J.Y., Hilscher, J. and Szilagyi, J., 2008. In search of distress risk. *The Journal of Finance*, *63*(6), pp.2899-2939.

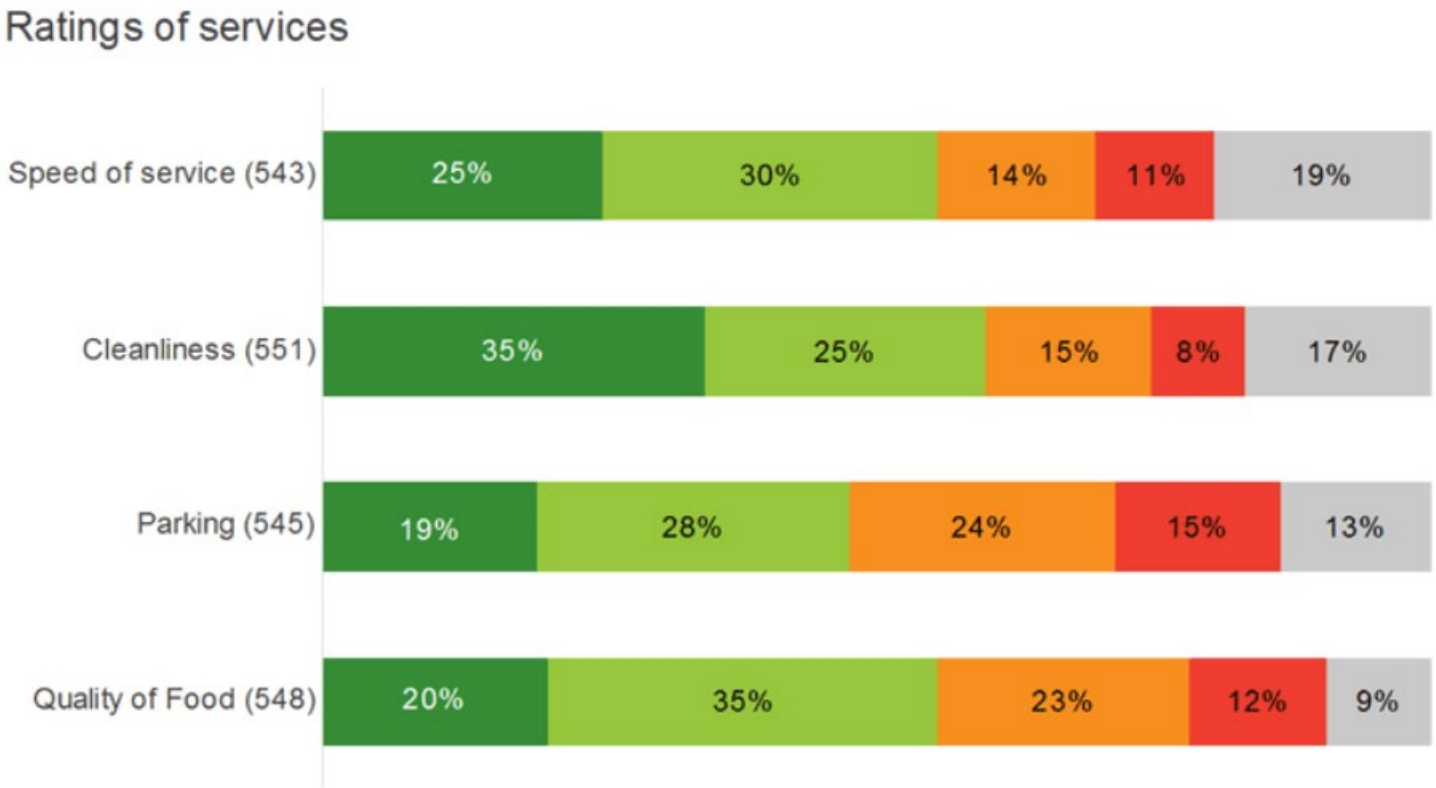
Panel A. Return Prediction

Characteristic Observed:	Beginning of Year	Beginning of Month
NIMTAAVG	14.55	38.99
	(4.09)**	(11.02)**
TLMTA	0.58	0.74
	(2.76)**	(3.65)**
EXRETAVG	11.50	0.88
	(6.78)**	(0.40)
SIGMA	−0.96	−2.05
	(2.68)**	(5.85)**
RSIZE	0.02	−0.02
	(0.48)	(0.45)
CASHMTA	2.93	4.33
	(8.49)**	(11.42)**
MB	−0.19	−0.05
	(6.61)**	(1.59)
PRICE	0.62	0.80
	(5.41)**	(6.64)**

In this regression, all the explanatory variables enter significantly with the exception of market capitalization. Of the significant variables, five enter with a sign opposite to that in the failure prediction regression—that is, they contribute to the low returns on distressed stocks—and two, leverage and stock price, enter with the same sign that they have in the failure prediction regression. Highly leveraged stocks are more likely to fail, but have delivered high average returns, while stocks with high prices are less likely to fail but have delivered low average returns. The former pattern may result from the salience of leverage to even unsophisticated investors, while the latter may reflect the value effect, imperfectly captured by the market-to-book ratio, or the size effect, imperfectly captured by our measure of relative market capitalization.

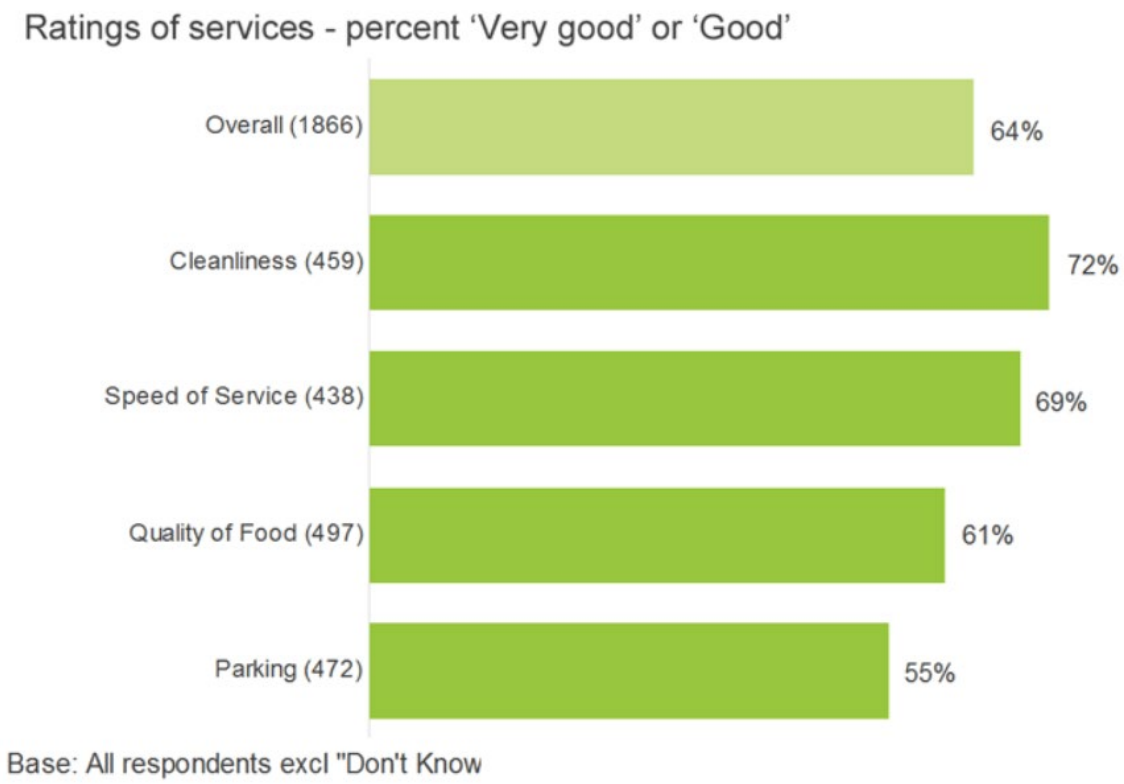
Survey Data Analysis

Visualization of sample



Survey Data Analysis

Visualization of sample



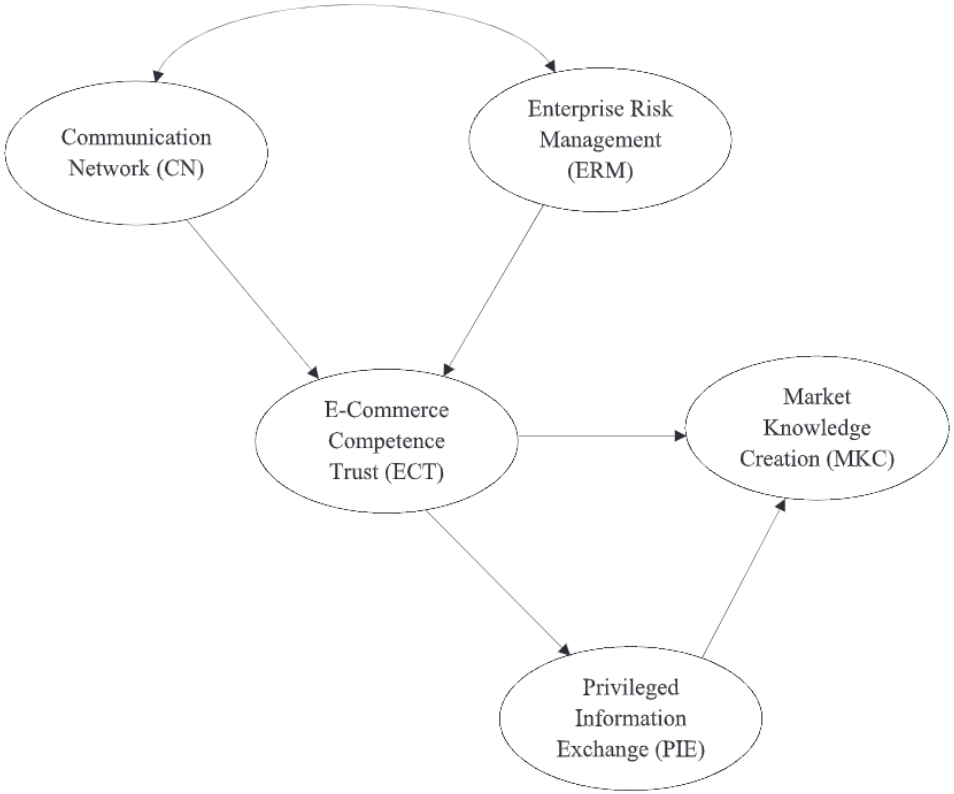
Survey Data Analysis

Word Clouds – a unique method of presenting textual data. Frequently occurring words are displayed with greater prominence or specific colors.
Content analysis – identify recurring themes and patterns in open-ended responses
Thematic analysis – group similar ideas and concepts together

Survey Data Analysis

Structural equation modeling (SEM): Investigate causal relationships among variables and understand how each contributes to overall performance.

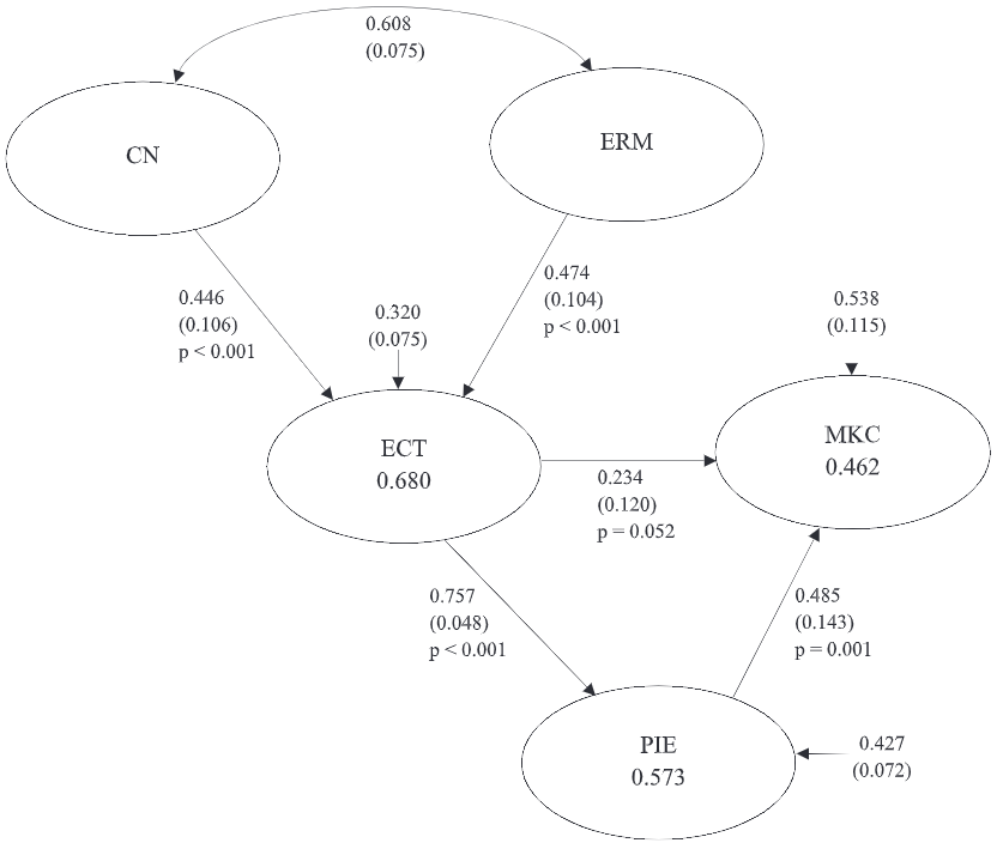
FIGURE 2
Proposed Structural Model



Survey Data Analysis

Structural equation modeling (SEM): Investigate causal relationships among variables and understand how each contributes to overall performance.

FIGURE 4
Structural Model Results for Model 1 (MLR Estimation)
(n = 197)[a,b]



Survey Data Analysis

Draw conclusions: Based on your analysis, what are the key findings and insights?

- The ECT to MKC path loading is not significant (0.234; p=0.052). All other paths loadings are strong, significant, and in the predicted direction.

Actionable insights: How can you use these findings to inform decision-making and improve your business or research?

Seminar Activity

- Continue with the previous paper that you've identified if there are Tables on summary statistics, correlation matrix and regression, identify the NEW variable(s) that is being proposed by the paper
- Find the summary statistics table, identify the discussion with a focus on the dependent variable (Y) and NEW variable(s)
- Find the correlation matrix table, identify the discussion with a focus on dependent variable (Y) and NEW variable(s)
- Find the first regression table, identify the discussion with a focus on dependent variable (Y) and NEW variable(s)
- Find the table on robustness check, identify the discussion with a focus on dependent variable (Y) and NEW variable(s)

Group Assignment (slides + data + code + present)

1. Find a paper that uses COMPUSTAT and CRSP data published after 2023 on top journals
2. Literature review & hypotheses development (pretend that you were the author)
   1. What did previous paper find
   2. Hypotheses: motivation and only ONE hypothesis from the paper
   3. Introduce a new variable from Refinitiv (or Bloomberg) and propose additional hypothesis on the new variable
3. Methodology: two regression models – one to test the main hypothesis in the original paper (H1), one to test your proposed hypothesis (H2)
4. Data (database, sample period, variable definitions, data cleansing) – each group member to download at least 10 companies (i.e., for a team of 5, you have 50 companies) since 2010.
   1. Merge CRSP, COMPUSTAT and Refinitiv variables
   2. Report the descriptive statistics (firm characteristics) and correlation matrix
5. Empirical results: main result from original table (H1), main result using your sample (H1), and one result for H2
6. Conclusion: three conclusions for each of the empirical results
7. Submission in week 9